Coffee Roast Model Card

roast-swin-retrained-v1

Intended use

Educational image-level roast inspection: Dark, Green (unroasted), Light, Medium. Owner: Group Idea, 6688218 Peerawat and 6688230 Yongyot. This release does not certify defects, SCA grades or individual beans.

Data and training

1194 unique 224 px RGB images; 835/178/181 train/validation/test; seed 42; zero exact-pixel duplicates. ImageNet Swin-T, 20 epochs, batch 8, AdamW and OneCycleLR. Best checkpoint: epoch 10 by validation loss. Training: 22.44 minutes on local Apple M1 MPS. Original Iteration4 baseline saw all original test images; its 99.75% result is historical.

Measured evaluation

Selected dynamic INT8 export: 29.60 MiB. Validation macro-F1:1.000 for both float and INT8; probability parity error 0.005897 on 9 validation inputs. INT8 median 64.3 ms versus float 130.2 ms on this Mac. Selected release evaluated once on 181 test images: accuracy 100.0%, macro-F1 1.000.

Test class	Support	Precision / recall / F1
Dark	45	1.000 / 1.000 / 1.000
Green	46	1.000 / 1.000 / 1.000
Light	45	1.000 / 1.000 / 1.000
Medium	45	1.000 / 1.000 / 1.000

Serving and limitations

RGB 224 bilinear resize and ImageNet normalization. JPEG/PNG only; 5 MiB and 20 MP limits. Mandatory SHA-256/class/preprocessing/version checks. Uniform camera frames reject 422 before inference. Source/camera confounding and near duplicates remain possible; arbitrary noncoffee photos can pass the guard. Confidence is uncalibrated. New-camera performance and group-separated validation are unmeasured.

License and lineage

Dataset: both version 1 Kaggle sources, gpiosenka/coffee-bean-dataset-resized-224-x-224 and sot2542/coffee-bean-dataset-v1, CC BY-SA 4.0. Attribution/modifications/citation: DATA_LICENSE.md. Dataset licensing does not automatically establish a weight license.

Git: 763915a0b9d05af803194239a3601a4a15dfa08f
DVC: d241ed318d2c1d8672d939938d3b71be.dir
ONNX SHA-256: 7d9949cbfe1177a28bd649f0e014c742bdac0941962b1a2a6cc1f4353702aafe
Azure ML: coffee-bean-roast-classifier version 1. Full parameters/dependencies/checkpoint hash are in the immutable release metadata.